

## AI/ML Virtual Try-On System – Model Comparison Report

### Objective

The objective of this study is to compare four widely used AI models—MediaPipe, OpenPose, Detectron2, and U2Net—in the context of virtual try-on applications. The comparison focuses on their functionality, accuracy, speed, ease of implementation, and suitability for body measurement extraction, pose estimation, and body segmentation tasks.

## 1. MediaPipe

### Overview

MediaPipe is an open-source framework developed by Google for building machine learning pipelines. It provides real-time pose estimation, face detection, hand tracking, and body landmark detection.

### Features

- Real-time body pose estimation
- Lightweight and fast execution
- 33 body landmarks detection
- Easy integration with Python applications
- Suitable for mobile and web platforms

### Advantages

- High processing speed
- Easy to implement
- Low computational requirements
- Good accuracy for real-time applications

### Limitations

- Less accurate for complex body poses
- Performance decreases with occlusions
- Limited customization compared to advanced frameworks



## 2. OpenPose

## Overview

OpenPose is a real-time multi-person pose estimation system developed by Carnegie Mellon University. It detects body, face, hand, and foot keypoints.

## Features

- Multi-person pose detection
- Detailed body keypoint estimation
- Face and hand landmark detection
- Widely used in research applications

## Advantages

- High pose estimation accuracy
- Supports multiple people simultaneously
- Robust landmark detection

## Limitations

- Requires high computational power
- Slower than MediaPipe
- More difficult to set up and deploy



## 3. Detectron2

### Overview

Detectron2 is a computer vision framework developed by Meta AI. It provides advanced object detection, instance segmentation, and keypoint detection capabilities.

### Features

- Object detection
- Instance segmentation
- Keypoint detection
- Flexible deep learning architecture

### Advantages

- High segmentation accuracy
- Powerful customization options
- Suitable for advanced computer vision projects



Limitations

- Complex installation process
- Higher hardware requirements
- Longer training and inference times

4. U2Net

Overview

U2Net is a deep learning model primarily designed for salient object detection and background removal. It is widely used for body segmentation tasks.

Features

- Accurate foreground extraction
- High-quality background removal
- Fine edge detection
- Lightweight architecture



Advantages

- Excellent segmentation quality
- Precise body boundary extraction
- Effective for virtual try-on preprocessing

Limitations

- Not suitable for pose detection
- Requires preprocessing for some applications
- Limited functionality outside segmentation tasks

Comparison Table

| Feature              | MediaPipe      | OpenPose       | Detectron2               | U2Net        |
|----------------------|----------------|----------------|--------------------------|--------------|
| Primary Function     | Pose Detection | Pose Detection | Segmentation & Detection | Segmentation |
| Accuracy             | High           | Very High      | Very High                | High         |
| Speed                | Very Fast      | Moderate       | Moderate                 | Fast         |
| Hardware Requirement | Low            | High           | High                     | Medium       |

| Feature                 | MediaPipe | OpenPose  | Detectron2 | U2Net          |
|-------------------------|-----------|-----------|------------|----------------|
| Ease of Implementation  | Easy      | Moderate  | Difficult  | Easy           |
| Real-Time Capability    | Excellent | Good      | Moderate   | Good           |
| Multi-Person Support    | Limited   | Excellent | Good       | Not Applicable |
| Segmentation Capability | Basic     | No        | Excellent  | Excellent      |

### Advantages and Limitations Summary

MediaPipe is best suited for real-time applications where speed and ease of implementation are important. OpenPose provides superior pose estimation accuracy and supports multiple people but requires higher computational resources. Detectron2 offers powerful object detection and segmentation capabilities, making it suitable for advanced computer vision systems. U2Net delivers excellent body segmentation and background removal, making it highly useful for virtual try-on preprocessing.

### Conclusion

Based on the comparison, MediaPipe is the most suitable choice for real-time body landmark detection due to its speed and simplicity. OpenPose provides more accurate pose estimation but requires stronger hardware. Detectron2 is ideal for advanced segmentation and object detection tasks, while U2Net excels in body segmentation and background removal.

For a complete virtual try-on system, a combination of MediaPipe for pose estimation and U2Net or Detectron2 for body segmentation can provide effective results. The selection of the model ultimately depends on project requirements, available computational resources, and desired accuracy.